XENON INTERPRETABILITY

Decision Structure Analysis

Can a linear probe, reading only the model's representation of an asset row, predict which asset the model will eventually trade? And does adding downstream context improve that prediction?

MODEL	TICKS	TRAIN / TEST	LAYERS	DATE
Qwen3-30B-A3B	918 (280 buy, 279 sell, 359 obs)	735 / 183	48	19 March 2026

KEY FINDING

The model already knows which asset it will trade from the market row alone.

Asset-binding is present in the earliest layers — AUROC 0.86–0.88 for buy targets at layer 0 — and adding downstream context (settings, portfolio, constraints) provides no meaningful lift. The decision target is encoded in the market representation before policy or portfolio information is available to the model.

Experiment Design

This experiment uses **real production prompts** — not synthetic counterfactuals — to probe whether the model's per-asset market representation already encodes which asset will be acted upon.

918 ticks were captured: 280 buys, 279 sells, and 359 observations. These were split 735 train / 183 test. For each tick, the model's hidden states were extracted at all 48 layers and pooled into per-asset row representations and per-section downstream representations.

For each captured tick, we have:

- **Market-only representations** — `row_mean` and `row_eos`: the model's activation at each asset row, pooled across tokens, before any downstream section has been processed.
- **Market + context representations** — `row_mean` concatenated with each downstream section's eos token: settings, portfolio, constraints, previous decisions, and last token.
- **Three probe targets** — `is_target_asset` (any traded asset), `is_buy_target` (bought asset), and `is_sell_target` (sold asset).

The probe is a grouped evaluation: for each tick (group), the 3–7 assets compete, and the probe must rank the actually-traded asset above the others. AUROC, Hit@1, and MRR are computed per-group and averaged. Test-set evaluation yields 114 groups for target asset (all trade ticks), 58 for buy targets (buy ticks only), and 56 for sell targets (sell ticks only).

What this tests

If the model encodes which asset it will act on before seeing settings and portfolio, it means the **market representation itself carries action-relevant preference**. If downstream context is needed, it means decision binding happens after policy integration.

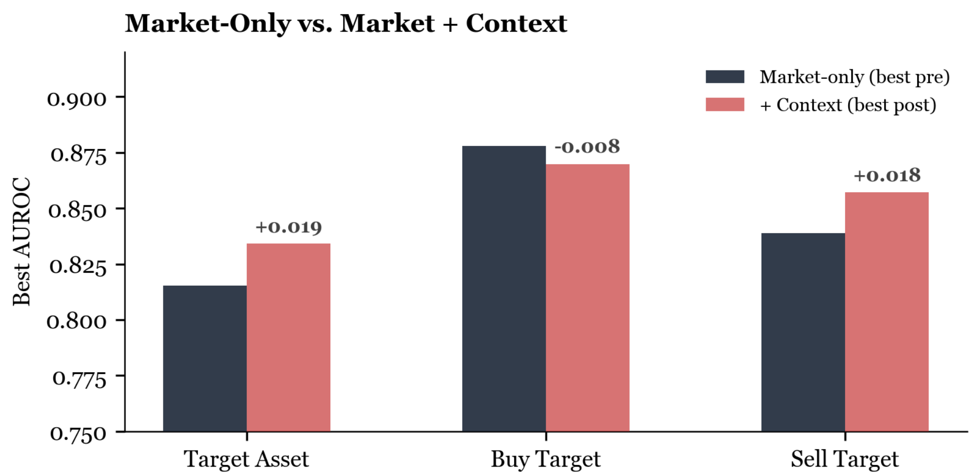
Summary

BUY TARGET (PRE)
0.878 AUROC

BUY TARGET (POST)
0.870 AUROC

SELL TARGET (PRE)
0.839 AUROC

SELL TARGET (POST)
0.857 AUROC



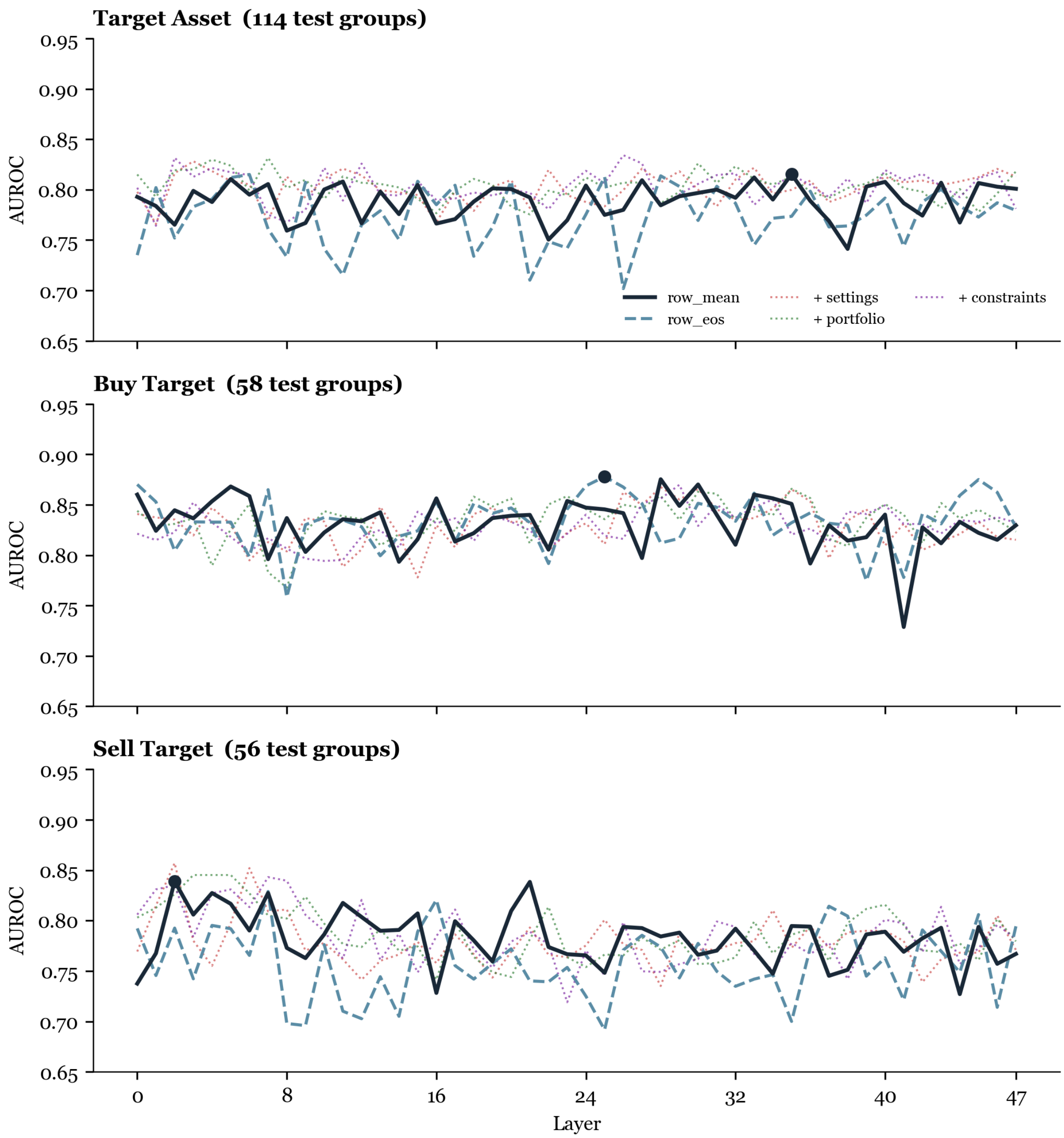
Target	Best Pre	Pre Rep.	Best Post	Post Rep.	Δ
is_target_asset	0.815 (L35)	row_mean	0.834 (L26)	1. constraints	+0.019
is_buy_target	0.878 (L25)	row_eos	0.870 (L29)	1. constraints	−0.008
is_sell_target	0.839 (L2)	row_mean	0.857 (L2)	1. settings	+0.018

INTERPRETATION

Adding downstream context provides **at most +0.019 AUROC** for any target — well within noise for n=56–114. For buy targets, context actually *hurts* (−0.008). The market representation is already sufficient.

is_target_asset — Which Asset Will Be Traded?

This target pools buys and sells: the probe must identify which of the 3–7 assets the model will act on, regardless of direction. 114 test groups.



Layerwise target-binding readout for all three targets. Solid navy = row_mean, dashed teal = row_eos, red = best post-context (+ constraints). All three targets show strong signal from the earliest layers.

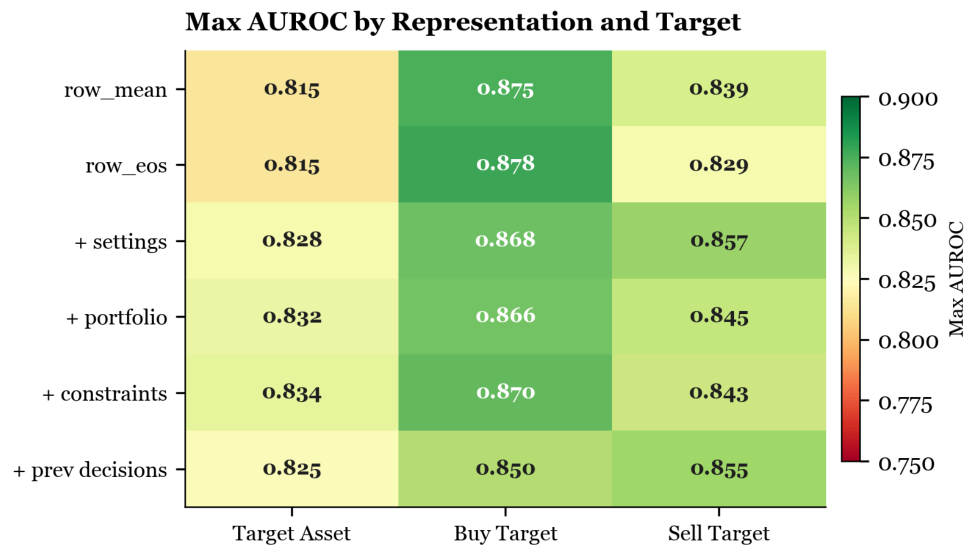
- **Layer 0 AUROC = 0.793.** The model's initial embedding already separates traded from non-traded assets well above chance.
- **No clear layer progression.** Performance oscillates between 0.75–0.82 without a clear upward trend. The signal is present early and doesn't sharpen much through the network.

- **Context representations track the same band.** Portfolio (`+portfolio_eos`) provides the most consistent lift, but all context variants stay within ±0.03 of market-only.

Representation	**Best AUROC**	**Layer**	**Hit@1**	**MRR**
row_mean	0.815	35	0.588	0.738
row_eos	0.815	6	0.553	0.712
row_mean + portfolio_eos	0.832	7	0.596	0.754
row_mean + constraints_eos	0.834	26	—	—
row_mean + settings_eos	0.828	3	—	—

is_buy_target — Which Asset Will Be Bought?

Restricted to the 58 test groups where the model chose BUY. The probe must identify which asset was bought. This is the strongest result.



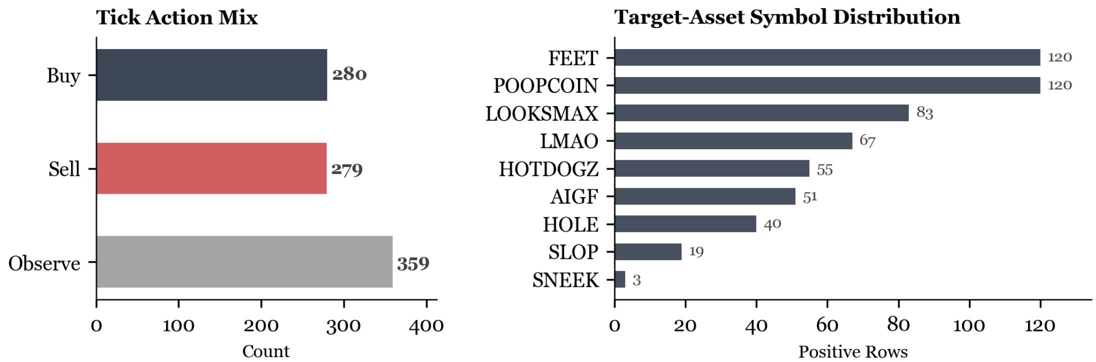
Max AUROC by representation and target. Market-only representations (row_mean, row_eos) match or exceed all context-augmented representations for buy targets.

> **KEY RESULT**
>
> **Buy targets are identifiable from market rows alone at 0.86–0.88 AUROC across most layers.** The model's representation of "which asset looks attractive for buying" is already present in the market section — before settings, portfolio, or any other context is processed. Adding context does not help; the best post-context AUROC (0.870) is slightly below the best pre-context (0.878).

- **Layer 0 AUROC = 0.860 (row_mean) and 0.870 (row_eos).** The embedding layer alone achieves nearly the best performance of any representation at any layer.
- **Peak: row_eos at L25 = 0.878.** The end-of-row token is slightly better than the row mean for buy targets, suggesting the model may accumulate a summary judgment at the last token of each asset block.
- **Hit@1 = 0.655 at best.** The probe correctly picks the exact bought asset 65% of the time — out of 6 candidates.
- **Context hurts.** Every market+context representation underperforms market-only. The downstream sections add noise to what is already a clean market signal.

is_sell_target — Which Asset Will Be Sold?

56 test groups where the model chose SELL. The probe must identify which held asset was sold.



Dataset composition: 280 buys, 279 sells, 359 observations across 918 ticks. Right panel shows target-asset symbol distribution — FEET and POOPCOIN dominate.

- **Sells are noisier than buys.** The AUROC curves are more volatile and the overall ceiling is lower (0.839 pre vs 0.878 for buys). This makes sense: selling involves portfolio-specific reasoning (unrealized PnL, time held, stop-loss thresholds) that isn't fully captured in market data alone.
- **Best pre: row_mean at L2 = 0.839.** Surprisingly, the best market-only performance comes at layer 2 — very early in the network. This suggests sell-target identity may be encoded via simple surface features.
- **Settings context provides the largest lift for sells: +0.018.** Adding `active_settings_eos` pushes to 0.857 at L2. This is the one target where context helps meaningfully, possibly because sell decisions are more policy-sensitive (sell rules, holding style, stop-loss thresholds).
- **Portfolio context helps early layers.** `row_mean+portfolio_eos` reaches 0.845 at L3, consistent with the idea that selling requires knowing what you hold.

What This Means

The model forms asset preferences during market reading

The most striking result is that buy-target identity is present at 0.86+ AUROC from **layer 0**. This is not a late-layer phenomenon. The model's initial encoding of market data already carries strong information about which asset will be acted upon. This suggests that **asset valence — bullish/ bearish preference — is a primary organizing principle of the market representation**, not something computed downstream.

Context adds noise, not signal, for buy decisions

For buy targets, every market+context representation underperforms market-only. This implies that buy decisions are fundamentally market-driven: the model identifies the most attractive asset from market data, and policy/portfolio context doesn't refine that choice. The buying decision is "which asset looks best?" — answered by the market alone.

Sell decisions are partially policy-dependent

Sells show modest context sensitivity. Settings (+0.018) and portfolio (+0.006) provide small lifts, consistent with sell decisions requiring awareness of holding style, stop-loss rules, and what's actually in the portfolio. But even here, the market-only baseline is strong (0.839), suggesting the model primarily identifies sell candidates from *market weakness*, then policy gates the execution.

prev_decisions_eos = last_token (confirmed bug)

> **KNOWN ISSUE**
>
> `row_mean+prev_decisions_eos` and `row_mean+last_token` produce **identical** values at every layer for every target. This confirms the boundary-detection aliasing bug flagged in the counterfactual report — both resolve to the same token position.

Connecting to the Research Plan

These results directly address hypotheses from the Market Manifold Research Plan:

- **H6 (Asset-Valence Manifold):** Confirmed. "Asset-specific bullishness is already visible in `row_mean_i` before settings." Buy-target AUROC of 0.86+ at layer 0 establishes this.
- **H7 (Decision Formation is Staged):** Partially confirmed. Target-asset binding happens before downstream context is integrated. But we need more targets (`act_vs_observe`, `size_bucket`) to establish the full staging order.
- **H4 (Post-settings transformation):** For buy decisions, settings add no signal. For sell decisions, settings add a small parallel lift. This is consistent with "policy overlay without rewriting market structure" — the orthogonal drift hypothesis.

Limitations and Next Steps

- **Test-set size:** 58 buy and 56 sell test groups (from 183 test ticks out of 918 total). The oscillation in per-layer AUROC (±0.05) likely reflects sampling variance from these group counts. Bootstrap CIs would help, but weren't computed in this run.
- **Label leakage risk:** If the model's market row encoding already incorporates information from the preamble (which precedes the market section in a causal model), the "market-only" representation is not purely market. The preamble may bias which assets look attractive. The counterfactual experiment's Q-A result (CKA ≥ 0.998) suggests this is minimal, but it was tested with a near-identical preamble.
- **Missing targets:** `act_vs_observe`, `size_bucket`, and `strategy_driven_vs_slider_driven` are needed to establish the full decision crystallization order (H7).
- **No regime stratification:** The research plan calls for splitting by constrained vs unconstrained, zero-ETH vs funded, etc. This analysis pools all ticks.
- **No causal tests:** Probes show decodability, not necessity. Activation patching or metric ablation would test whether the model actually *uses* the market-row signal vs just correlating with it.
- **Row randomization control missing:** If the same asset tends to appear at the same position across captures, the probe might learn position rather than asset identity. Need to verify row order was randomized.

Immediate next steps from the research plan

1. **Intrinsic dimensionality** (Measurement 1) — how many effective dimensions does the market row representation use?
2. **RSA against hand-crafted feature matrices** (Measurement 2) — does the model organize assets by raw metrics, ranks, or derived factors?
3. **Per-metric decodability curves** (Measurement 3) — which market metrics drive the buy/sell signal?
4. **Pairwise relational encoding** (Measurement 4) — does the model compare assets relationally?
5. **Fix the boundary parser** to resolve the `prev_decisions_eos` / `last_token` aliasing.

Generated from experiment `decision_structure` on 19 March 2026. 918 production ticks (280 buy, 279 sell, 359 observe), 735 train / 183 test. Qwen3-30B-A3B surrogate with vLLM hidden state capture across 48 layers.